Transport in animals

30 questions: 15 multiple choice, 15 structured.

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1 · Part 1

Core Circulatory systems [1]

The circulatory system is made of blood vessels, a pump and valves.

Which part of the circulatory system ensures that blood flows in one direction only?

A the thick walls of the arteries

B the valves in the heart and veins

C the thin walls of the capillaries

D the platelets in the blood

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2 · Part 1

Extended Circulatory systems [1]

How many times does blood pass through the heart during one complete circuit of the body in a fish and in a mammal?

	fish	mammal
A	twice	once
B	once	once
C	twice	twice
D	once	twice

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3 · Part 1 Core Heart [1]

Which statement about the direction of blood flow in blood vessels is correct?

A Arteries carry blood away from the heart and veins carry blood back to the heart.

B Arteries carry blood back to the heart and veins carry blood away from the heart.

C Arteries and veins both carry blood away from the heart.

D Veins carry blood away from the heart and capillaries carry it back to the heart.

4 · Part 1 Core Heart [1]

Which list gives three ways of monitoring the activity of the heart?

A breathing rate, ECG, pulse rate

B ECG, body temperature, listening to the sounds of valves closing

C ECG, pulse rate, listening to the sounds of valves closing

D pulse rate, breathing rate, body temperature

5 · Part 1 Core Heart [1]

The table shows the heart rate of a student before and immediately after running for two minutes.

condition	heart rate / beats per minute
at rest	72
immediately after running	126

What is the percentage increase in the student's heart rate?

A 75%

B 54%

C 43%

D 175%

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6 · Part 1 Core Heart [1]

Which statement describes coronary heart disease?

A The vena cava becomes blocked, so less blood returns to the heart.

B The coronary arteries become blocked, so the heart muscle receives less oxygen and glucose.

C The valves of the heart are damaged, so blood flows backwards.

D The pulmonary artery becomes blocked, so less blood reaches the lungs.

7 · Part 1 Extended Heart [1]

Why is the muscle wall of the left ventricle thicker than the muscle wall of the right ventricle?

A The left ventricle pumps a much larger volume of blood than the right ventricle.

B The left ventricle pumps blood to the lungs, which needs a very high pressure.

C The left ventricle receives blood from the vena cava at a high pressure.

D The left ventricle pumps blood at a higher pressure, all around the body.

8 · Part 1 Extended Heart [1]

The muscles of the ventricles contract and blood is forced out of the heart.

Which row shows the state of the valves while blood is leaving the ventricles?

	atrioventricular valves	semilunar valves
A	open	open
B	open	closed
C	closed	open
D	closed	closed

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9 · Part 1 Core Blood vessels [1]

Which row describes the structure of a vein?

	thickness of wall	diameter of lumen	valves
A	thick	narrow	absent
B	thin	wide	present
C	thin	wide	absent
D	thick	wide	present

10 · Part 1 Core Blood vessels [1]

Blood flows through the capillaries of a leg muscle.

Which exchange takes place between the blood and the muscle cells?

A Carbon dioxide moves from the blood into the muscle cells and oxygen moves from the muscle cells into the blood.

B Glucose moves from the muscle cells into the blood and carbon dioxide moves from the blood into the muscle cells.

C Oxygen and glucose move from the blood into the muscle cells and carbon dioxide moves from the muscle cells into the blood.

D Red blood cells move from the blood into the muscle cells to deliver oxygen.

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11 · Part 1 Extended Blood vessels [1]

Veins contain valves but arteries do not.

Which statement explains why veins need valves?

A Blood in veins is at high pressure, so valves are needed to slow it down.

B Veins have thin walls, so valves are needed to stop the walls from bursting.

C Valves contract to pump blood along the veins towards the heart.

D Blood in veins is at low pressure, so valves are needed to prevent backflow.

12 · Part 1 Core Blood [1]

Which component of blood transports urea, hormones and carbon dioxide?

A plasma

B platelets

C red blood cells

D white blood cells

13 · Part 1 Core Blood [1]

Which statement describes how red blood cells transport oxygen?

A They engulf oxygen molecules by phagocytosis.

B They use oxygen to release energy for the rest of the body.

C They contain antibodies that attach to oxygen.

D They contain haemoglobin, which combines with oxygen.

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14 · Part 1 Extended Blood [1]

Which row shows the functions of lymphocytes and phagocytes?

	lymphocytes	**phagocytes**
A	engulf pathogens by phagocytosis	produce antibodies
B	produce antibodies	engulf pathogens by phagocytosis
C	produce antibodies	produce antibodies
D	form blood clots	engulf pathogens by phagocytosis

15 · Part 1 Extended Blood [1]

Which statement describes the process of blood clotting?

A Fibrin is converted to fibrinogen, which forms a mesh that traps blood cells.

B Platelets are converted to fibrinogen, which forms a mesh that traps blood cells.

C Fibrinogen is converted to fibrin, which forms a mesh that traps blood cells.

D Fibrinogen is converted to fibrin, which digests pathogens entering the wound.

16 · Part 1 Core Circulatory systems [4]

The circulatory system of a mammal is a system of blood vessels with a pump and valves.

(a) Name the organ that acts as the pump. [1]

(b) State the function of the valves in the circulatory system. [1]

(c) One valve in a patient's circulatory system is damaged and does not close fully.

Suggest two effects this could have on the flow of blood. [2]

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17 · Part 1 Extended Circulatory systems [6]

Fish have a single circulation. Mammals have a double circulation.

(a) Describe the single circulation of a fish. [2]

(b) Describe the double circulation of a mammal. [3]

(c) State which side of a mammal's heart receives blood that has just passed through the lungs. [1]

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18 · Part 1 Extended Circulatory systems [6]

Scientists measured the mean blood pressure in two blood vessels of a fish and in two blood vessels of a mammal.

animal	blood vessel	mean blood pressure / kPa
fish	vessel carrying blood from the heart to the gills	4.0
fish	vessel carrying blood from the gills to the rest of the body	2.4
mammal	aorta	12.0
mammal	pulmonary artery	2.0

(a) Calculate the percentage decrease in mean blood pressure as blood passes through the gills of the fish. [2]

(b) Using the information in the table, explain the advantages of a double circulation for a mammal. [4]

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19 · Part 1 Core Heart [6]

Some students investigated the effect of physical activity on heart rate. One student sat still for five minutes and then measured her pulse rate. It was 72 beats per minute.

She then did step-ups onto a bench for three minutes. She measured her pulse rate immediately after stopping and then every minute for six minutes.

time after stopping exercise / min	pulse rate / beats per minute
0	132
1	118
2	101
3	88
4	78
5	74
6	74

(a) Describe how the student could measure her pulse rate. [1]

(b) Calculate the increase in pulse rate between rest and immediately after the exercise. [1]

(c) Describe the changes in pulse rate after the exercise stopped. Use data from the table in your answer. [2]

(d) The student wants to compare her results with those of a friend.

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Suggest two variables that should be kept the same. [2]

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20 · Part 1 Core Heart [8]

Coronary heart disease (CHD) is a common cause of death in many countries.

(a) Describe what happens in coronary heart disease. [2]

(b) State two risk factors for coronary heart disease that a person cannot change. [2]

(c) Explain how diet and exercise can reduce the risk of coronary heart disease. [4]

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21 · Part 1 Extended Heart [6]

Some students cut open a sheep's heart and measured the thickness of the muscle walls of the two ventricles.

chamber	thickness of muscle wall / mm
left ventricle	14
right ventricle	4

(a) Calculate the ratio of the thickness of the left ventricle wall to the thickness of the right ventricle wall. Give your answer in the form x : 1. [1]

(b) Explain why the muscle wall of the left ventricle is thicker than the muscle wall of the right ventricle. [3]

(c) Explain why the muscle walls of the atria are thinner than the muscle walls of the ventricles. [2]

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22 · Part 1 Extended Heart [6]

Deoxygenated blood from the vena cava enters the right atrium of the heart.

(a) Describe how this blood is moved from the right atrium into the pulmonary artery. In your answer, refer to the contraction of the muscles and the action of the valves. [5]

(b) Doctors can listen to the heart using a stethoscope.

State what causes the sounds that are heard. [1]

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23 · Part 1 Extended Heart [6]

The heart rate of a cyclist was measured during different activities.

activity	heart rate / beats per minute
sitting	64
cycling slowly	96
cycling quickly	138
sprinting uphill	182

(a) Explain why the heart rate increases as the activity becomes more intense. [4]

(b) After sprinting uphill, the cyclist's heart rate stays high for several minutes even though she has stopped cycling.

Explain why. [2]

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24 · Part 1 Extended Heart [6]

Some babies are born with a hole in the septum between the left and right ventricles. These babies often tire easily.

(a) State the function of the septum. [1]

(b) Explain why a hole in the septum reduces the amount of oxygen delivered to the baby's body tissues. [3]

(c) State two ways in which a doctor could monitor the activity of the baby's heart. [2]

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25 · Part 1 Core Blood vessels [6]

The table gives information about three blood vessels, X, Y and Z, from the arm of a person.

blood vessel	thickness of wall / mm	diameter of lumen / mm	valves
X	1.0	4.0	absent
Y	0.5	5.0	present
Z	0.001	0.008	absent

(a) Identify the type of blood vessel X, Y and Z. [2]

(b) Give one piece of evidence from the table which shows that Y is a vein. [1]

(c) Calculate how many times wider the lumen of Y is than the lumen of Z. [1]

(d) State two functions of capillaries. [2]

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26 · Part 1 Extended Blood vessels [6]

The blood pressure in an artery in the arm rises and falls with each heartbeat, between 10.7 kPa and 16.0 kPa. The blood pressure in a nearby vein stays at about 1.0 kPa.

(a) Explain how the structure of the artery is related to the pressure of the blood inside it. [3]

(b) Explain how the structure of the vein is related to the pressure of the blood inside it. [3]

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27 · Part 1 Extended Blood vessels [6]

Capillaries carry blood through the tissues of the body. Their walls are only one cell thick. The lumen of a capillary is so narrow that red blood cells pass through it in single file.

(a) Explain how the thickness of the capillary wall is related to the function of capillaries. [2]

(b) Explain how the narrow lumen helps the exchange of oxygen between the blood and the tissues. [2]

(c) Capillaries form large, highly branched networks in tissues such as muscle.

Explain how this helps the tissue cells. [2]

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28 · Part 1 Core Blood [8]

Blood is made of several components.

(a) List the four components of blood. [2]

(b) State the function of platelets. [1]

(c) State two functions of white blood cells. [2]

(d) State the role of haemoglobin in red blood cells. [1]

(e) Name two substances, other than urea, that are transported in the plasma. [2]

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29 · Part 1 Core Blood [6]

(a) Someone cuts their finger with a knife. Blood flows out and then a clot forms.

(i) State two roles of blood clotting. [2]

(ii) Name the component of blood involved in clotting. [1]

(b) Some people have far fewer red blood cells in their blood than normal. These people often feel tired and have little energy.

Explain why. [3]

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30 · Part 1 Extended Blood [6]

Bacteria enter the body through a cut in the skin.

(a) Describe the process of blood clotting at the cut. [2]

(b) State the function of:

(i) phagocytes [1]

(ii) lymphocytes. [1]

(c) Some bacteria entered the tissue around the cut before the clot formed. Scientists counted the number of phagocytes in samples of tissue from around the cut on several days.

time after cut / days	number of phagocytes in sample
0	2
1	15
2	34
3	28
5	9

Suggest reasons for the changes in the number of phagocytes. [2]

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